Amir Golparvar

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PROFESSIONAL SUMMARY

Berlin-based full-stack developer with 6+ years of experience building and deploying production FastAPI + React applications end-to-end. Own the full stack from database schema and API design to polished UIs, and deploy to both single-node (Docker Compose) and distributed cloud environments (AWS, Kubernetes, Helm). Ship with TDD/BDD using pytest, pytest-bdd, and Playwright, backed by thorough CI/CD pipelines (GitLab CI, GitHub Actions). Experienced with MongoDB, PostgreSQL, Elasticsearch, OAuth/OIDC, Stripe billing, and distributed task processing with Temporal and Celery.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, C++
Frontend	React, Vite, Material UI (MUI), Plotly.js, TanStack Query, Lexical, Storybook
Backend	FastAPI, Pydantic, Node.js, Express.js
Databases	MongoDB, PostgreSQL, Elasticsearch
Authentication	OAuth 2.0, OpenID Connect (OIDC), JWT, Keycloak
DevOps & CI/CD	Docker, Docker Compose, Kubernetes, Helm, nginx, GitHub Actions, GitLab CI/CD, AWS EC2, DigitalOcean
Testing	pytest, pytest-bdd, Playwright, unit testing, integration testing, E2E testing
Messaging & Workers	Celery, Temporal, RabbitMQ, Apache Kafka
Other	Stripe, RAG/LLM integration, MkDocs, Postman, Git

WORK EXPERIENCE

Full-Stack Developer

Apr 2022 – Present

FAIRmat, Humboldt-Universität zu Berlin — Berlin, Germany

- Build and operate FastAPI microservices for data ingestion, validation, and processing with Celery/Temporal workers, powering a platform that manages 19M+ entries and 114 TB+ of materials science data
- Deploy and maintain NOMAD Oasis instances on single-node servers (Docker Compose) and distributed cloud infrastructure (AWS EC2, Kubernetes + Helm), orchestrating Elasticsearch, MongoDB, RabbitMQ, Keycloak, and nginx behind reverse proxy
- Develop and maintain the NOMAD GUI (React + Vite + MUI + TanStack Query + Plotly): uploads, faceted search UX, data viewers, and admin tools serving a 600+ researcher community across 10+ institutions
- Build reusable React components for data visualization and API response caching, and develop NOMAD plugins (parsers, schema packages) supporting 60+ scientific file formats
- Architect internal RAG assistant ("NOMAD Compass") that surfaces documentation, schemas, and code examples, reducing onboarding friction for new users
- Co-develop the Electronic Lab Notebook (ELN) module and third-party integrations enabling researchers to capture and share experimental workflows
- Drive test-driven development practices: end-to-end (Playwright), integration, and unit tests with pytest and pytest-bdd; document modules using MkDocs and Storybook

Backend Developer

Dec 2020 – Mar 2022

Freelance / Personal Projects — Germany

- Designed and built a RESTful API (Node.js, Express, MongoDB) for managing bootcamp websites with full CRUD operations, deployed to Amazon EC2 and DigitalOcean
- Implemented JWT and cookie-based authentication with encrypted passwords, reset tokens, and security headers (Helmet), preventing NoSQL injection attacks
- Designed producer/consumer messaging microservices using Apache Kafka (Docker) and RabbitMQ for asynchronous event processing

Frontend Developer

Dec 2020 – Mar 2022

Freelance / Personal Projects — Germany

- Built responsive websites and landing pages using React (class-based and functional component architectures) and Node.js
- Created a scientific web application that loads micro-CT images, applies configurable thresholds, and renders interactive cross-sectional slices

Scientific Researcher

Oct 2018 – Oct 2022

Helmholtz Centre for Environmental Research (UFZ) — Leipzig, Germany

- Developed P3D-BRNS, an open-source C++/OpenFOAM numerical toolbox for simulating multiphase, multicomponent reactive transport in micro-CT images of porous structures
- Published 6 peer-reviewed papers and presented findings at 3 international conferences (EGU, CMWR, InterPore)

Graduate Teaching Assistant

Jan 2017 – Oct 2018

University of Aberdeen — Aberdeen, United Kingdom

- Taught Reservoir Simulation (postgraduate), Computational Fluid Dynamics (postgraduate), Engineering Mathematics and Fluid Mechanics (undergraduate)

EDUCATION

PhD, Geoscience & Microbiology

Leibniz Universität Hannover — Hannover, Germany | *Jul 2021 – Present*

Research Fellow, Petroleum Engineering

University of Aberdeen — Aberdeen, United Kingdom | *Jan 2016 – Oct 2018*

MSc, Petroleum Reservoir Engineering

Shiraz University — Shiraz, Iran | *Nov 2012 – Feb 2015*

BSc, Petroleum Engineering

Petroleum University of Technology — Ahwaz, Iran | *Aug 2008 – Apr 2012*

PUBLICATIONS

- **2024** — P3D-BRNS v1.0.0: A three-dimensional, multiphase, multicomponent, pore-scale reactive transport modelling package for simulating biogeochemical processes in subsurface environments
- **2023** — NOMAD: A distributed web-based platform for managing materials science research data
- **2021** — Pore-scale modeling of microbial activity: What we have and what we need — *Vadose Zone Journal*
- **2019** — A comprehensive review of pore scale modeling methodologies for multiphase flow in porous media — *AGER*
- **2018** — A support vector machine analysis to predict density of mixtures of methanol and six ionic liquids — *Monatshefte für Chemie*
- **2014** — A Critical Review of Improved Oil Recovery by Electromagnetic Heating

LANGUAGES

English (C1 – Proficient) | German (A2 – Basic)